

CS-671

Deep Learning Hackathon

PS: Spoofing your own Anti-Spoof System

Group No. – 27

Group Members:

1. B24241
2. B24238
3. B24123
4. B24125
5. B24268
6. B24316
7. B24045
8. B24126
9. B24139
10. B24118

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1 Project Overview

This report documents the full experimental pipeline for the voice anti-spoofing challenge. The work is organized into four distinct tasks:

- **Task 1** — Binary classifier development to distinguish bonafide from spoofed speech, evaluated on both in-domain and cross-domain splits.
- **Task 2** — Voice cloning and similarity analysis utilizing ECAPA-TDNN embeddings.
- **Task 3** — Stress testing the selected classifier under additive noise (AWGN and Babble) at varying Signal-to-Noise Ratio (SNR) levels.
- **Task 4** — System integration into an interactive demonstration for real-time prediction.

2 Dataset and Experimental Protocol

The **ASVspoof 2019 dataset** is used for training and evaluation. This dataset contains three types of attacks: Text-to-Speech (TTS), Voice Conversion (VC), and Replay. The dataset is categorized into two distinct partitions:

- **Logical Access (LA)**
- **Physical Access (PA)**

Set A (LA) and Set B (PA) are utilized with strictly disjoint splits. **Test data was not used** for training, model selection, or threshold tuning at any stage.

Experimental Runs:

- **Run 1:** Train/Validate on Set A (LA) → Evaluate on Set A (in-domain) and Set B (cross-domain).
 - **Run 2:** Train/Validate on Set B (PA) → Evaluate on Set B (in-domain) and Set A (cross-domain).
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3 Task 1 — Anti-Spoofing Classifier

3.1 RawNet and SpecRNet

3.1.1 *RawNet*

RawNet operates directly on raw audio waveforms, typically utilizing 1D convolutional layers, followed by recurrent layers (e.g., GRU) to capture temporal representations.

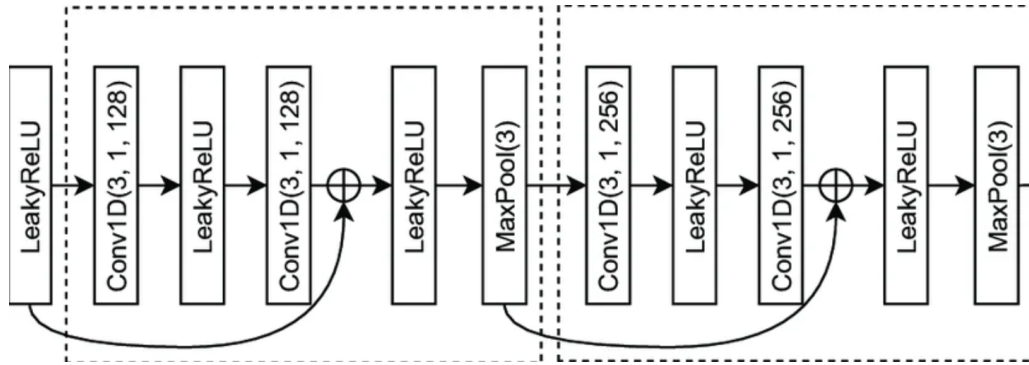


Figure 1: RawNet Architecture.

3.1.2 *SpecRNet*

SpecRNet extracts representations from 2D spectral features (e.g., LFCCs or Mel-spectrograms). It utilizes 2D convolutional blocks with residual connections.

Table 1: Primary evaluation metrics for RawNet and SpecRNet.

Model	In-dom. EER	In-dom. Acc	Cross-dom. EER	Cross-dom. Acc
RawNet	0.1959	0.5647	0.3552	0.1591
SpecRNet	0.2627	0.6025	0.4600	0.2227

Conclusion: Due to sub-optimal preliminary results, the baseline CNN models were dropped from further extensive testing.

3.2 Audio Mamba

Audio Mamba leverages selective state space models (SSMs) to process long audio sequences with linear time complexity.

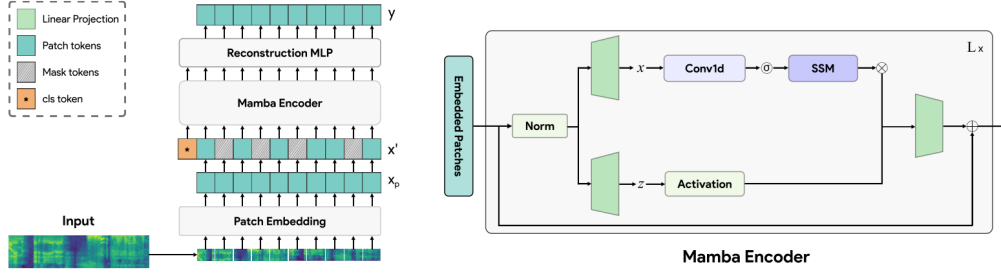


Figure 2: Audio Mamba Architecture.

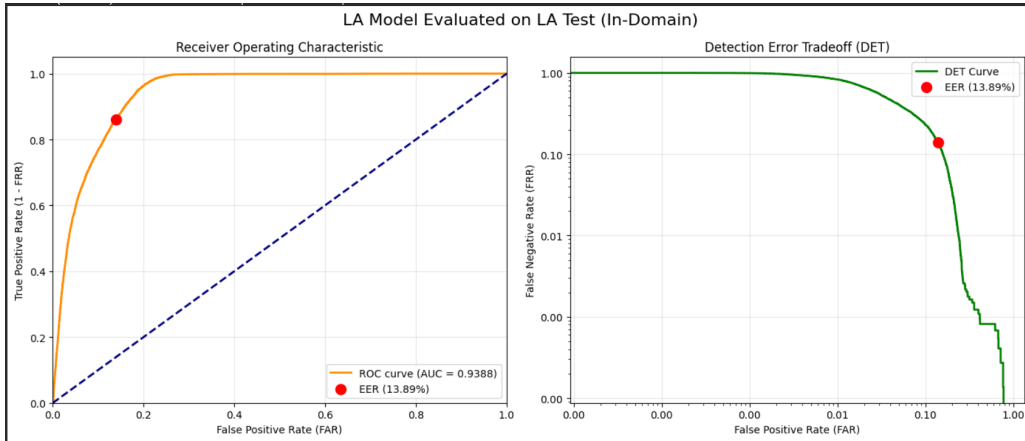


Figure 3: Training and validation curves for Audio Mamba (In-Domain).

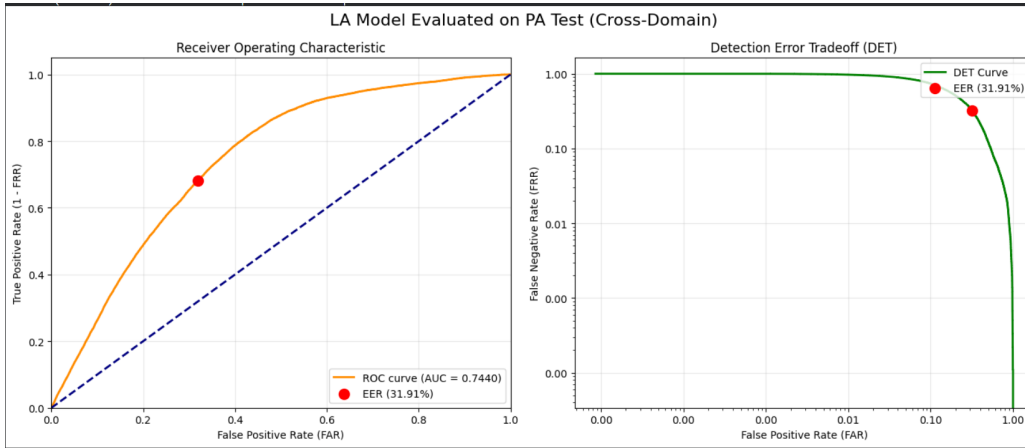


Figure 4: Training and validation curves for Audio Mamba (Cross-Domain).

Conclusion: Because the DET Curves of Audio Mamba exhibited counter-productive characteristics (high false rejection rates) despite having better raw accuracy, the model was dropped from production consideration.

3.3 AASIST (Primary Model)

3.3.1 Model Architecture

AASIST employs an integrated end-to-end architecture featuring a **SincConv front-end** (out_channels=70, kernel_size=1025) coupled with a **Residual Encoder** containing 6 ResBlocks with MaxPool1d, LeakyReLU, and residual skip connections. This is followed by a heterogeneous graph attention network (Graph-Attentional layer) and a max-graph operation to model both temporal and spectral artifacts directly from the raw waveform.

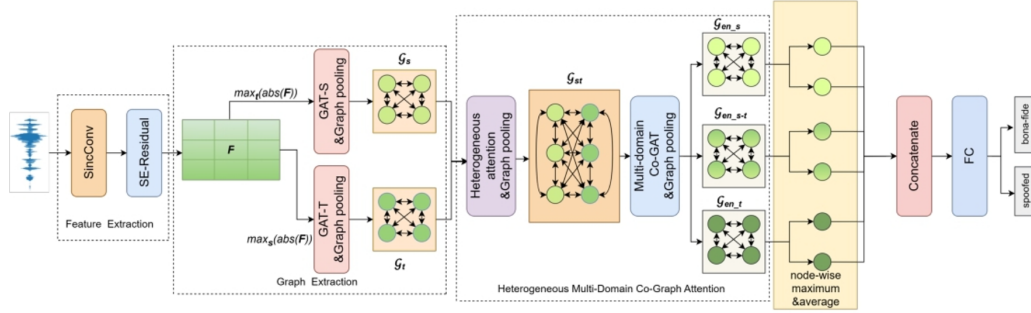


Figure 5: AASIST Architecture Diagram.

3.3.2 Pre-processing

The following pre-processing steps are rigidly applied to the data pipeline:

- **Waveform Loading and Resampling** — Converts all audio to a fixed 16 kHz sampling rate for consistency.
- **Pre-emphasis Filtering** — Enhances high-frequency components to highlight synthetic speech artifacts.
- **Energy-based VAD** — Removes silent/low-energy regions to keep only vital speech segments.
- **RMS Normalization** — Standardizes loudness dynamically across all audio samples.
- **Padding / Cropping** — Ensures fixed-length inputs (64,600 samples) for efficient batch training.

3.3.3 Training Diagnostics and Hyperparameters

Table 2: Training hyperparameters for AASIST (Run 1 and Run 2).

Hyperparameter	Value
Optimizer	Adam
Initial Learning Rate	0.0001
Weight Decay	0.0001
LR Scheduler	CosineAnnealingLR (T_max=20, eta_min=1e-06)
Loss Function	Weighted CrossEntropyLoss
Batch Size	64
Early Stopping Criteria	patience=6, min_delta=1e-4 (monitor: val EER)
Epochs Trained (Max)	20
Gradient Clipping	max_norm=1.0
Input Length (Samples)	64,600
Sample Rate	16,000 Hz
Automatic Mixed Precision	True
Workers	4
Random Seed	42

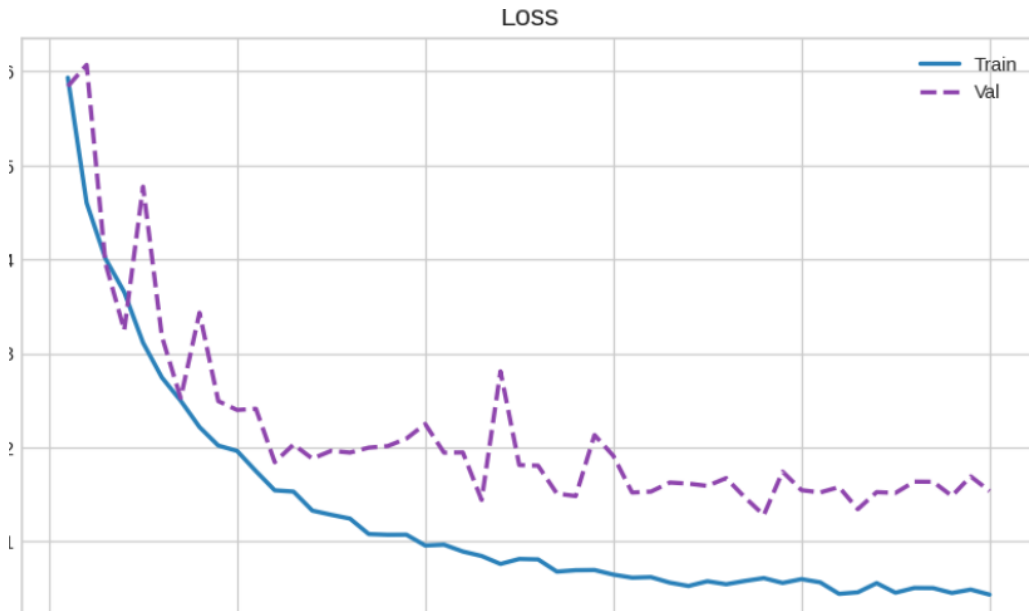


Figure 6: Training and validation loss curves for AASIST (Run 1).

3.3.4 Model Complexity

Table 3: Model complexity for AASIST.

Metric	Run 1 (Model A)	Run 2 (Model B)
Total Trainable Parameters	104,666	104,666
Saved Model Size (MB)	0.44	0.44

3.3.5 Performance Metrics

Table 4: EER and AUC summary results for AASIST.

Evaluation Condition	Run 1 (Model A)		Run 2 (Model B)	
	EER (%)	ROC AUC	EER (%)	ROC AUC
In-domain	13.89	0.9388	25.77	0.8192
Cross-domain	31.91	0.7440	36.44	0.6953

Table 5: Detailed classification metrics for AASIST. Decision threshold derived from validation set at EER operating point.

Metric	In-domain				Cross-domain			
	Run 1 (Model A)		Run 2 (Model B)		Run 1 (Model A)		Run 2 (Model B)	
	Bona Fide	Spoof	Bona Fide	Spoof	Bona Fide	Spoof	Bona Fide	Spoof
Accuracy (%)	79.56		72.84		56.77		42.92	
CM [TN,FP,FN,TP]	[943, 22, 2277, 8007]		[1468, 422, 4052, 10528]		[1739, 151, 6969, 7611]		[821, 144, 6277, 4007]	
Precision	0.2929	0.9973	0.2659	0.9615	0.1997	0.9805	0.1157	0.9653
Recall	0.9772	0.7786	0.7767	0.7221	0.9201	0.5220	0.8508	0.3896
F1-score	0.4507	0.8745	0.3962	0.8248	0.3282	0.6813	0.2036	0.5552

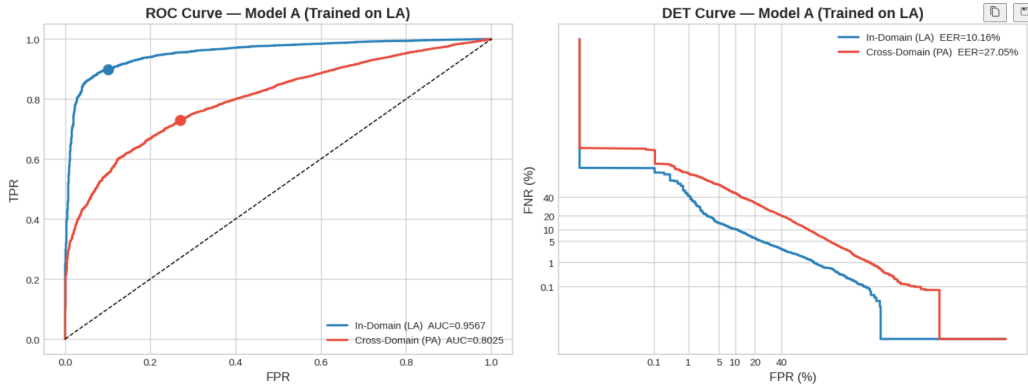


Figure 7: ROC and DET curves for AASIST (Run 1).

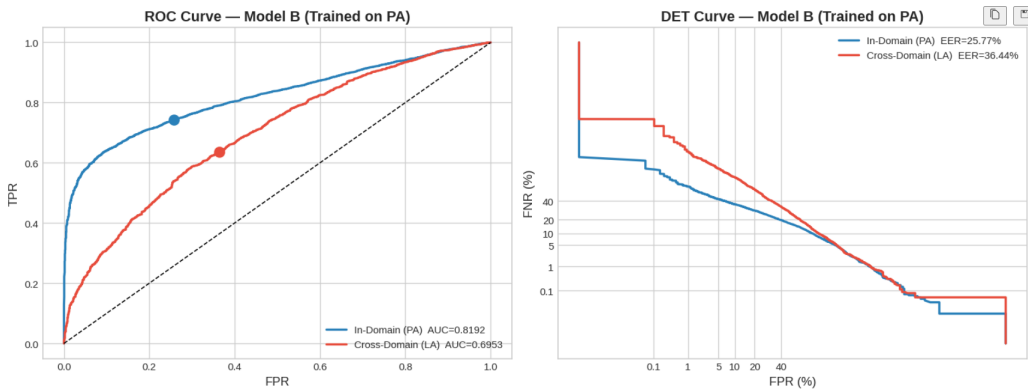


Figure 8: ROC and DET curves for AASIST (Run 2).

3.3.6 *Benchmarking and Computation Profile*

All benchmarks use a single arbitrary 5-second audio clip as input (16,000 Hz). Inference latency is the mean over 100 independent iterations.

Table 6: Computational cost and inference latency (5-second input).

Metric	Run 1 (Model A)	Run 2 (Model B)
MACs	530.202 M	530.202 M
FLOPs (approx)	1060.404 M	1060.404 M
Mean Latency (ms)	3.882	3.853

4 Task 2 — Voice Cloning and Similarity Analysis

Voice cloning was performed using **XTTS**. For each of the two speakers, 25 unique sentences were recorded (real) and then cloned (synthetic), yielding 100 total samples: 50 real and 50 synthetic, evenly split across the two speakers. Each real utterance is directly paired with its corresponding synthetic counterpart (same sentence, same speaker).

Audio format standard: 16 kHz, mono, PCM-16.

4.1 Speaker Recording Details

Table 7: Speaker metadata.

Speaker ID	Gender	Sentences Recorded	Total Duration (s)
Speaker 1	Male	25	125
Speaker 2	Male	25	125

4.2 Cosine Similarity (ECAPA-TDNN)

Cosine similarity is computed between each real utterance and its corresponding synthetic counterpart using ECAPA-TDNN speaker embeddings.

Table 8: Cosine similarity between real and synthetic utterances.

Speaker	Mean Cosine Similarity
Speaker 1	0.4779
Speaker 2	0.5401
Overall Average	0.5090

4.3 Plots

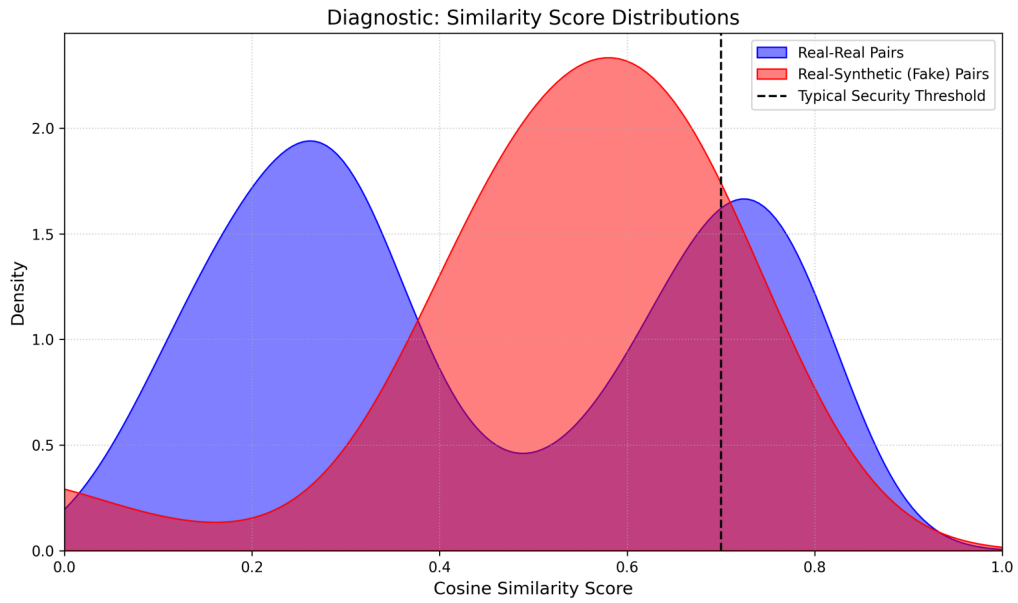


Figure 9: Similarity Distribution Plots.

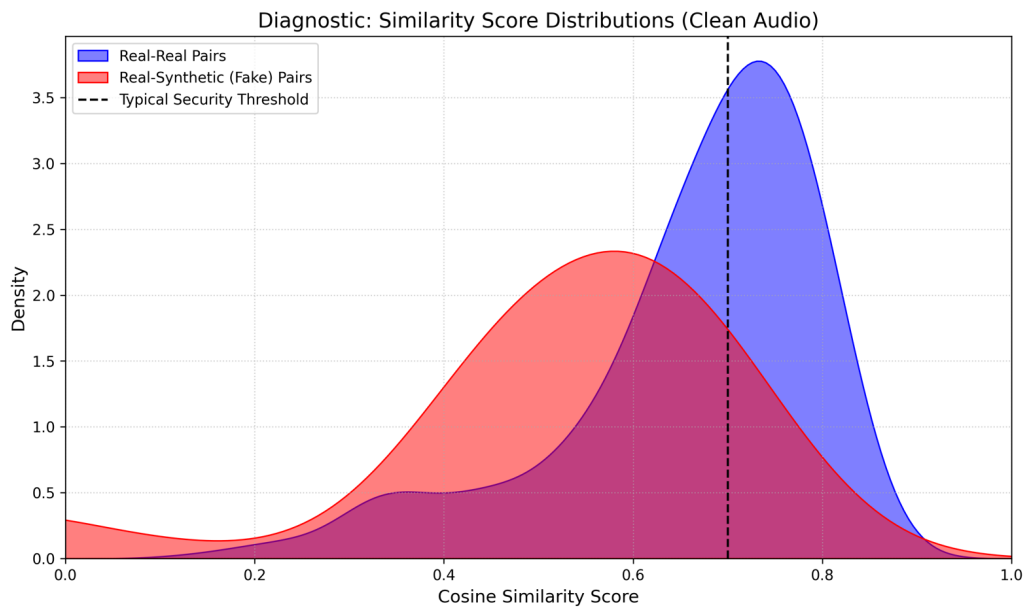


Figure 10: Cosine Similarity Scores.

4.4 Attack Success Metrics

The 100-sample Task 2 dataset (50 real, 50 synthetic) was evaluated using both Task 1 classifiers with their fixed decision thresholds.

Table 9: Attack success metrics on the Task 2 dataset.

Metric	Run 1 (Model A)	Run 2 (Model B)
EER (%)	20.00	52.00
Attack Success Rate (%)	40.00	64.00
TN / FP / FN / TP	[46, 4, 20, 30]	[28, 22, 32, 18]
F1-score (Spoofed)	0.7143	0.4000

5 Task 3 — Stress Testing

Both Run 1 and Run 2 classifiers were evaluated on the Task 2 dataset under six noise conditions formed by crossing two noise types with three SNR levels. **No retraining or threshold adjustment was performed**; all thresholds are fixed from Task 1 validation.

Noise types: Additive White Gaussian Noise (AWGN) and Babble noise.

SNR levels: 20 dB, 10 dB, and 5 dB.

Noise intensity is dynamically scaled per clip using the clean signal’s RMS energy to reach the target SNR prior to addition.

5.1 Baseline Performance (Clean)

Table 10: Classifier performance on clean Task 2 samples.

Metric	Run 1 (Model A)	Run 2 (Model B)
EER (%)	20.00	52.00
Bypass Rate (%)	40.00	64.00
TN / FP / FN / TP	[46, 4, 20, 30]	[28, 22, 32, 18]

5.2 Results Under Noise — Run 1

Table 11: Run 1 (Model A) classifier performance across all noise conditions.

Noise Type	SNR (dB)	EER (%)	Bypass Rate (%)	TN / FP / FN / TP
AWGN	20	0.00	0.00	[42, 8, 0, 50]
AWGN	10	0.00	0.00	[41, 9, 0, 50]
AWGN	5	0.00	0.00	[41, 9, 0, 50]
Babble	20	31.00	42.00	[47, 3, 21, 29]
Babble	10	35.00	46.00	[48, 2, 23, 27]
Babble	5	39.00	56.00	[48, 2, 28, 22]

5.3 Results Under Noise — Run 2

Table 12: Run 2 (Model B) classifier performance across all noise conditions.

Noise Type	SNR (dB)	EER (%)	Bypass Rate (%)	TN / FP / FN / TP
AWGN	20	62.00	82.00	[26, 24, 41, 9]
AWGN	10	81.00	80.00	[13, 37, 40, 10]
AWGN	5	82.00	42.00	[1, 49, 21, 29]
Babble	20	56.00	70.00	[28, 22, 35, 15]
Babble	10	59.00	76.00	[28, 22, 38, 12]
Babble	5	60.00	80.00	[29, 21, 40, 10]

6 Reproducibility and Hardware

All results reported in this document are strictly reproducible. Running the codebase with the exact configuration will yield deterministic results.

Fixed random seeds:

Table 13: Random seeds used across all experiments.

Library	Seed Value
NumPy	42
PyTorch	42

Table 14: Hardware and software environment used for all experiments.

Component	Specification
CPU	AMD EPYC 7313 16-Core Processor
GPU	NVIDIA RTX A5000 24GB VRAM
Operating System	Linux
Python Version	3.12.3

7 Results Summary

Table 15: Consolidated results across all tasks.

Task	Metric	Run 1	Run 2
Task 1 In-domain	EER (%)	13.89	25.77
Task 1 Cross-domain	EER (%)	31.91	36.44
Task 2	Attack EER (%)	20.00	52.00
Task 2	Avg Cosine Similarity	0.5090	
Task 3 Worst Case	EER (%) at AWGN 5 dB	0.00	82.00
Task 3 Worst Case	Bypass Rate (%) at AWGN 5 dB	0.00	42.00

8 References

- [1] **AASIST**: <https://ieeexplore.ieee.org/stamp/stamp.jsp?arnumber=9747766>
- [2] **Audio Mamba**: <https://ieeexplore.ieee.org/stamp/stamp.jsp?arnumber=9747766>
- [3] **ASVspoof 2019**: [https://www.kaggle.com/datasets/awsaf49/asvspoof-2019-dataset?resource=dow](https://www.kaggle.com/datasets/awsaf49/asvspoof-2019-dataset?resource=download)